



Student number: _____

GIRRAWEEEN HIGH SCHOOL

2022

TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Extension 1

**General
Instructions**

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided
- In section II, show relevant mathematical reasoning and/or calculations

**Total marks:
70****Section I – 10 marks**

- Attempt Questions 1-10
- Allow about 15 minutes for this section

Section II – 60 marks

- Attempt all questions
- Allow about 1 hour and 45 minutes for this section

Section I

10 marks

Attempt questions 1 - 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for questions 1-10

1. Given that α and β are both acute angles, evaluate $\sin(\alpha + \beta)$ if

$$\sin \alpha = \frac{8}{17} \text{ and } \sin \beta = \frac{4}{5}.$$

- A. $\frac{108}{85}$ B. $\frac{84}{85}$ C. $\frac{36}{85}$ D. $\frac{28}{85}$

2. If $\vec{u} = 2\vec{i} + 3\vec{j}$, the unit vector in the direction of $\vec{u}$ is:

- A. $\frac{1}{13}(2\vec{i} + 3\vec{j})$
B. $\frac{1}{5}(2\vec{i} + 3\vec{j})$
C. $\sqrt{13}(2\vec{i} + 3\vec{j})$
D. $\frac{1}{\sqrt{13}}(2\vec{i} + 3\vec{j})$

3. A polynomial $P(x)$ has a triple root at $x = -2$.

Which of the following statements is true?

- A. $(x + 2)^2$ is a factor of $P'(x)$.
B. $(x - 2)^2$ is a factor of $P'(x)$.
C. $(x + 2)^3$ is a factor of $P'(x)$.
D. $(x - 2)^3$ is a factor of $P'(x)$.

4. Which of the following is an expression for $\int \sin^2 4x \, dx$?

A. $\frac{x}{2} - \frac{1}{16} \sin 8x + c$

B. $\frac{x}{2} + \frac{1}{16} \sin 8x + c$

C. $x - \frac{1}{8} \sin 8x + c$

D. $x + \frac{1}{8} \sin 8x + c$

5. The equation $x^3 - 2x^2 - 4x + 8 = 0$ has roots α , β and γ .

What is the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$?

A. $-\frac{1}{2}$

B. $-\frac{1}{4}$

C. $\frac{1}{4}$

D. $\frac{1}{2}$

6. The domain and range of $y = 4 \cos^{-1} \left(\frac{3x}{2} \right)$ is

A. Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$

Range: $-2\pi \leq y \leq 2\pi$

B. Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$

Range: $0 \leq y \leq 4\pi$

C. Domain: $-\frac{2}{3} \leq x \leq \frac{2}{3}$

Range: $0 \leq y \leq \frac{\pi}{4}$

D. Domain: $-\frac{3}{2} \leq x \leq \frac{3}{2}$

Range: $0 \leq y \leq 4\pi$

7. $\int \sec^2 \theta \tan^2 \theta \, d\theta$ is
- A. $\sec^2 \theta + \frac{1}{2} \tan^2 \theta + c$
- B. $\frac{1}{3} \tan^3 \theta + c$
- C. $\frac{1}{3} \tan^3 \theta + \frac{1}{5} \tan^5 \theta + c$
- D. $\tan^4 \theta - \ln |\cos^4 \theta| + c$
8. A committee of 5 people is to be chosen from a group of 6 girls and 4 boys. How many different committees could be formed that have at least one boy?
- A. $\binom{10}{5} - 1$
- B. $\binom{4}{1} \times \binom{6}{4}$
- C. $\binom{4}{1} + \binom{6}{4}$
- D. $\binom{10}{5} - 6$
9. Given that n students are to be put into 4 classrooms, what is the minimum number of students to guarantee that there are at least two students born in the same month in each of the 4 classrooms?
- A. 48 B. 49 C. 52 D. 53
10. Given the parametric equations $\begin{cases} x = \sin^{-1} t + 1 \\ y = \frac{1}{\sqrt{1-t^2}} \end{cases}$
- A. $y = \frac{1}{1 - \sin(x-1)}$
- B. $y = \frac{1}{\sqrt{2 \sin x - \sin^2 x}}$
- C. $y = \frac{1}{\cos(x-1)}$
- D. $y = \pm \frac{1}{\cos(x-1)}$

Section II

60 marks

Attempt all questions

Allow about 1 hour and 45 minutes for this section

Start each question on a new page in the answer booklet provided.

Your responses should include relevant mathematical reasoning and/or calculations.

Extra writing space is available on request.

Question 11 (14 marks)

a. Solve $\frac{3}{3x-1} \leq 5$ [3]

b. Evaluate $\lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{3}}{4x} \right)$ [1]

c. Using the substitution $u = \frac{1}{x}$, find the exact value of

$$\int_1^2 \frac{e^{\frac{1}{x}}}{x^2} dx$$
 [3]

d. Evaluate $\int_0^{\sqrt{5}} \frac{2x^3}{\sqrt{x^2+4}} dx$ using the substitution $u = x^2 + 4$ [4]

e. Use sums to products of trigonometric ratios to solve the equation $\cos x + \cos 2x + \cos 3x = 0, 0 \leq x \leq 2\pi$ [3]

Question 12 (12 marks)

- a. (i) Express $3 \cos x + 2 \sin x$ in the form $R \cos(x - \alpha)$,
where $R > 0$ and $0^\circ < \alpha < 90^\circ$ [3]
- (ii) Hence, solve $3 \cos x + 2 \sin x = \sqrt{13}$, $0^\circ \leq x \leq 360^\circ$.
Give your answer in degrees correct to two decimal places. [2]
- b. Prove by mathematical induction that
 $2^{2n} + 6n - 1$ is divisible by 3 for all integers $n \geq 1$. [3]
- c. Consider the function $f(x) = \sqrt{2x - 1} + 1$
- (i) Find the equation of the inverse function $f^{-1}(x)$, stating its domain. [2]
- (ii) Find the x - coordinates of the points where the graphs of
 $y = f(x)$ and $y = f^{-1}$ intersect. [2]

Question 13 (14 marks)

- a. Find $\int_0^{\frac{4}{3}} \frac{dx}{16 + 9x^2}$ [3]
- b. (i) Show that $\int_0^{\frac{3\pi}{2}} \sin^2 x \, dx = \frac{3\pi}{4}$ [2]
- (ii) The graph of $y = 1 + \sin x$ for $0 \leq x \leq \frac{3\pi}{2}$ is rotated
about the x - axis. Find the exact volume of the solid generated. [2]
- c. A class of fifteen Mathematics students are seated around a circular table to discuss
Mathematical problems.
- (i) In how many different ways can the students be arranged? [1]
- (ii) If the seats are randomly assigned, find the probability that four particular
students, Will, Mike, Dustin and Lucas are not all seated together as a
group of four? [2]
- d. Consider the polynomial $P(x) = 4x^3 - 12x^2 + 5x + 6$
- (i) Use the factor theorem to show that $(x - 2)$ is a factor of $P(x)$. [1]
- (ii) Factorise $P(x)$ completely. [3]

Question 14 (10 marks)

a. Given that $\underline{a} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}$ and $\underline{b} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$, find

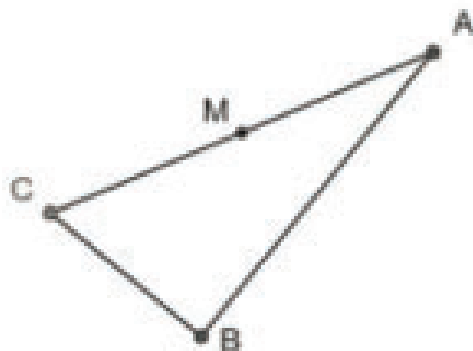
(i) $|\underline{a}|$ [1]

(ii) $\underline{a} \cdot \underline{b}$ [1]

(iii) The exact angle between $\underline{a}$ and $\underline{b}$ [2]

(iv) $\text{proj}_{\underline{b}} \underline{a}$ [2]

b. ΔABC is a right-angled triangle with M being the midpoint of the hypotenuse AC , as shown below. Let $\overrightarrow{AM} = \underline{a}$ and $\overrightarrow{BM} = \underline{b}$.



(i) Find $\overrightarrow{AB}$ and $\overrightarrow{BC}$ in terms of $\underline{a}$ and $\underline{b}$. [2]

(ii) Prove that M is equidistant from the three vertices of ΔABC . [2]

Question 15 (10 marks)

- a. A heated metal ball is dropped into a liquid.

As the ball cools, its temperature, $T^{\circ}\text{C}$, t minutes after it enters the liquid is given by:

$$T = 25 + 400 e^{-0.05t}, t \geq 0$$

- (i) Find the temperature of the ball as it enters the liquid. [1]

- (ii) Find the value of t when $T = 300$.
Answer correct to 3 significant figures. [1]

- (iii) Find the rate at which the temperature of the ball is decreasing at the instant when $t = 50$.
Give your answer in $^{\circ}\text{C}$ per minute to 3 significant figures. [2]

- (iv) Using the equation given above, explain why the temperature of the ball can never fall to 20°C [1]

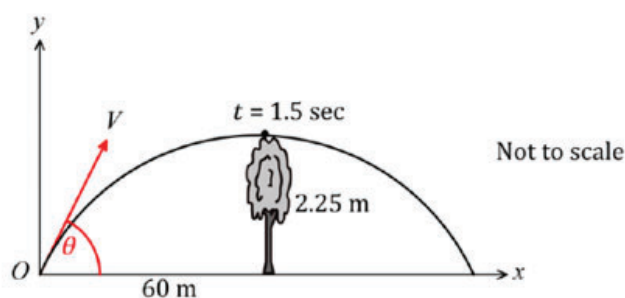
- b. A golfer hits a golf ball from a point O with speed V metres per second

at an angle of θ above the horizontal, where $0 \leq \theta \leq \frac{\pi}{2}$.

The ball just clears a 2.25 metre high tree after 1.5 seconds.

The tree is 60 metres away from the point from which the ball was hit.

Assume $g = 10\text{ms}^{-2}$.



- (i) Show that the horizontal and vertical displacement of the ball are given by
 $x = Vt \cos \theta$ and $y = -5t^2 + Vt \sin \theta$. [2]

- (ii) Find the angle of projection of the ball to the nearest minute. [2]

- (iii) Find the initial speed of the ball. [1]

End of Examination

①

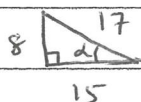
2022 GHS EXT 1 TRIAL SOLUTIONS

MC

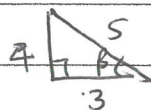
① B ② D ③ A ④ A ⑤ D ⑥ B ⑦ B ⑧ D ⑨ C ⑩ C

1. $\sin \alpha = \frac{8}{17}$

$\sin \beta = \frac{4}{5}$



$\cos \alpha = \frac{15}{17}$



$\cos \beta = \frac{3}{5}$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$= \frac{8}{17} \times \frac{3}{5} + \frac{15}{17} \times \frac{4}{5}$$

$$= \frac{84}{85}$$

[B]

2. $\hat{u} = \frac{u}{|u|}$

$$= \frac{2i - 3j}{\sqrt{2^2 + 3^2}}$$

$$= \frac{1}{\sqrt{13}} (2i + 3j)$$

[D]

3. $P(x) = (x+2)^3 Q(x)$

$$P'(x) = 3(x+2)^2 Q(x) + (x+2)^3 Q'(x)$$

$$= (x+2)^2 [3Q(x) + (x+2)Q'(x)]$$

$$\therefore (x+2)^2 \text{ is a factor of } P'(x). \quad [A]$$

4. $\int \sin^2 4x \, dx = \frac{1}{2} \int (1 - \cos 8x) \, dx$

$$= \frac{1}{2} \left(x - \frac{1}{8} \sin 8x \right) + C$$

$$= \frac{x}{2} - \frac{1}{16} \sin 8x + C$$

[A]

(2)

$$5. x^3 - 2x^2 - 4x + 8 = 0$$

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$$

$$= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma}$$

$$= \frac{c/a}{-d/a}$$

$$= \frac{-4}{-8}$$

$$= \frac{1}{2} \quad [D]$$

$$6. D: -1 \leq \frac{3x}{2} \leq 1$$

$$-\frac{2}{3} \leq x \leq \frac{2}{3}$$

$$R: 0 \leq \frac{y}{4} \leq \pi$$

$$0 \leq y \leq 4\pi \quad [B]$$

$$7. \int \sec^2 \theta \tan^2 \theta d\theta$$

$$u = \tan \theta$$

$$du = \sec^2 \theta d\theta$$

$$= \int u^2 du$$

$$= \frac{u^3}{3} + C$$

$$= \frac{1}{3} \tan^3 \theta + C \quad [B]$$

(3)

8. At least 1 boy = Total - No boys

$$= {}^{10}C_5 - {}^6C_5$$

$$= \binom{10}{5} - 6 \quad [D]$$

9. worst case scenario $\rightarrow$ there are 12 students all
 born in different months in each room $\Rightarrow 48$
 One more student in each room will guarantee
 that there are at least 2 students in each
 classroom who are born in the same month.
 $\therefore$ A minimum of 52 students required.

[C]

$$10. \begin{cases} x = \sin^{-1} t + 1 \\ y = \frac{1}{\sqrt{1-t^2}} \end{cases} \Rightarrow x-1 = \sin^{-1} t$$

$$\sin(x-1) = t$$

$$\therefore y = \frac{1}{\sqrt{1-(\sin(x-1))^2}}$$

$$= \frac{1}{\sqrt{\cos^2(x-1)}}$$

$$= \frac{1}{\cos(x-1)} \quad [C]$$

Section II

Question 11 (14 marks)

$$a) \frac{3}{3x-1} \leq 5$$

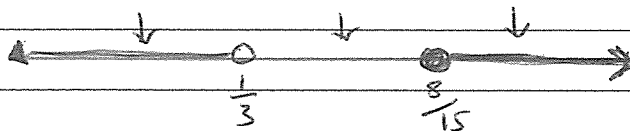
$$x \neq \frac{1}{3}$$

$$\text{Solve } \frac{3}{3x-1} = 5$$

$$5(3x-1) = 3$$

$$3x-1 = \frac{3}{5}$$

$$x = \frac{8}{15}$$



$$\text{Test } x=0$$

$$\frac{3}{-1} \leq 5$$

✓

$$\text{Test } x = \frac{1}{2}$$

$$\frac{3}{1/2} \leq 5$$

$$6 \leq 5$$

✗

$$\text{Test } x=1$$

$$\frac{3}{2} \leq 5$$

✓

$$x < \frac{1}{3} ; x \geq \frac{8}{15}$$

$$b) \lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{3}}{4x} \right)$$

$$= \lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{3}}{12(\frac{x}{3})} \right)$$

$$= \frac{1}{12} \lim_{x \rightarrow 0} \left(\frac{\sin \frac{x}{3}}{\frac{x}{3}} \right)$$

$$= \frac{1}{12}$$

5

$$11 \quad c) \int_1^2 \frac{e^{\frac{1}{x^2}}}{x^2} dx$$

$$u = \frac{1}{x^2} = x^{-2}$$

$$= - \int_1^2 e^u du$$

$$\frac{du}{dx} = -2x^{-3} = -\frac{1}{x^2}$$

$$du = -\frac{1}{x^2} dx$$

$$= - \left[e^u \right]_1^2$$

$$\text{when } x=2, u = \frac{1}{4}$$

$$x=1, u=1$$

$$= e^{-\frac{1}{4}} - e^{-1}$$

$$d) \int_0^{\sqrt{5}} \frac{2x^3}{\sqrt{x^2+4}} dx$$

$$u = x^2 + 4 \Rightarrow x^2 = u - 4$$

$$du = 2x dx$$

$$= \int_4^9 \frac{u-4}{\sqrt{u}} du$$

$$\text{when } x=0, u=4$$

$$x=\sqrt{5}, u=9$$

$$= \int_4^9 \left(u^{\frac{1}{2}} - 4u^{-\frac{1}{2}} \right) du$$

$$= \left[\frac{2}{3} u^{\frac{3}{2}} - 8u^{\frac{1}{2}} \right]_4^9$$

$$= \left(\frac{2}{3} (9)^{\frac{3}{2}} - 8(9)^{\frac{1}{2}} \right) - \left(\frac{2}{3} (4)^{\frac{3}{2}} - 8(4)^{\frac{1}{2}} \right)$$

$$= \frac{14}{3}$$

⑥

$$11 \text{ e) } \cos x + \cos 2x + \cos 3x = 0 ; \quad 0 \leq x \leq 2\pi$$

$$\left(\begin{array}{l} \text{using } \cos A \cos B = \frac{1}{2} [\cos (A-B) + \cos (A+B)] \\ \cos x + \cos 3x = 2 \cos 2x \cos x \end{array} \right)$$

$$\therefore \cos x + \cos 2x + \cos 3x = 0$$

$$\Rightarrow \cos 2x + 2 \cos 2x \cos x = 0$$

$$\cos 2x (1 + 2 \cos x) = 0 \quad (0 \leq 2x \leq 2\pi)$$

$$\cos 2x = 0$$

$$\cos x = -\frac{1}{2}$$

$$2x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}$$

$$x = \frac{2\pi}{3}, \frac{4\pi}{3}$$

$$\therefore x = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}, \frac{2\pi}{3}, \frac{4\pi}{3}$$

(7)

Question 12

$$\begin{aligned} \text{a) i) } 3 \cos x + 2 \sin x &= R \cos(x - \alpha) \\ &= R \cos x \cos \alpha + R \sin x \sin \alpha \end{aligned}$$

Equating coefficients,

$$R \cos \alpha = 3 \text{ --- (1)} ; R \sin \alpha = 2 \text{ --- (2)}$$

Squaring and adding (1) and (2)

$$R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 9 + 4$$

$$R^2 (\cos^2 \alpha + \sin^2 \alpha) = 13$$

$$R = \sqrt{13} , R > 0$$

$$\therefore \cos \alpha = \frac{3}{\sqrt{13}} , \sin \alpha = \frac{2}{\sqrt{13}} \quad \therefore 0^\circ < \alpha < 90^\circ$$

$$\alpha = 33.69^\circ \text{ (to 2dp)}$$

$$\therefore 3 \cos x + 2 \sin x = \sqrt{13} \cos(x - 33.69^\circ)$$

$$\text{(ii) } 3 \cos x + 2 \sin x = \sqrt{13}$$

$$\Rightarrow \sqrt{13} \cos(x - 33.69^\circ) = \sqrt{13}$$

$$\cos(x - 33.69^\circ) = 1$$

$$-33.69^\circ \leq x - 33.69^\circ \leq 326.31^\circ$$

$$\therefore x - 33.69^\circ = 0^\circ$$

$$\therefore x = 33.69^\circ$$

(8)

12 b) $2^{2n} + 6n - 1$ is divisible by 3, $n \geq 1$

Step 1 : Show true for $n=1$

$$2^{2(1)} + 6(1) - 1$$

= 9 which is divisible by 3

Step 2 : Assume true for $n=k$

$$\text{i.e. } 2^{2k} + 6k - 1 = 3P \quad \text{where } P \text{ is an integer.}$$

$$2^{2k} = 3P - 6k + 1$$

Step 3 : Prove true for $n=k+1$

$$\text{i.e. } 2^{2(k+1)} + 6(k+1) - 1 = 3Q \quad \text{where } Q \text{ is an integer.}$$

$$\text{LHS} = 2^{2(k+1)} + 6(k+1) - 1$$

$$= 2^{2k+2} + 6k + 6 - 1$$

$$= 2^2 \cdot 2^{2k} + 6k + 5$$

$$= 4(3P - 6k + 1) + 6k + 5 \quad (\text{from assumption})$$

$$= 12P - 24k + 4 + 6k + 5$$

$$= 12P - 18k + 9$$

$$= 3(4P - 6k + 3)$$

$$= 3Q$$

$$= \text{RHS}$$

$\therefore$ If the result is true for $n=k$, then
also true for $n=k+1$

Step 4 : By the principle of mathematical induction, the result is true for
all integers $n \geq 1$.

(9)

$$12 \text{ c) } f(x) = \sqrt{2x-1} + 1$$

$$\left(\begin{array}{l} D: x \geq \frac{1}{2} \\ R: y \geq 1 \end{array} \right)$$

$$i) f(x) : y = \sqrt{2x-1} + 1$$

$$f^{-1}(x) : x = \sqrt{2y-1} + 1$$

$$(x-1)^2 = 2y-1$$

$$2y = (x-1)^2 + 1$$

$$y = \frac{1}{2} [(x-1)^2 + 1]$$

$$\text{Domain : } x \geq 1$$

ii) At the points of intersection

$$\sqrt{2x-1} + 1 = x$$

$$2x-1 = (x-1)^2$$

$$2x-1 = x^2 - 2x + 1$$

$$x^2 - 4x + 2 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{4 \pm \sqrt{4^2 - 4(1)(2)}}{2}$$

$$= \frac{4 \pm \sqrt{8}}{2}$$

$$= 2 \pm \sqrt{2}$$

$\therefore$ x-coordinate of point of intersection

$$x = 2 + \sqrt{2} \quad (\text{since } x \geq 1)$$

Question 13 (14 marks)

$$a) \int_0^{4/3} \frac{dx}{16 + 9x^2}$$

$$= \frac{1}{3} \int_0^{4/3} \frac{3 dx}{4^2 + (3x)^2}$$

$$= \frac{1}{3} \times \frac{1}{4} \left[\tan^{-1} \frac{3x}{4} \right]_0^{4/3}$$

$$= \frac{1}{12} \left[\tan^{-1}(1) - \tan^{-1}(0) \right]$$

$$= \frac{\pi}{48}$$

$$b) (i) \int_0^{3\pi/2} \sin^2 x \, dx$$

$$= \frac{1}{2} \int_0^{3\pi/2} (1 - \cos 2x) \, dx$$

$$= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{3\pi/2}$$

$$= \frac{1}{2} \left[\left(\frac{3\pi}{2} - \frac{1}{2} \sin 3\pi \right) - 0 \right]$$

$$= \frac{3\pi}{4}$$

(11)

b (ii)

$$V = \pi \int_0^{3\pi/2} (1 + \sin x)^2 dx$$

$$= \pi \int_0^{3\pi/2} (1 + 2\sin x + \sin^2 x) dx$$

$$= \pi \left[\int_0^{3\pi/2} (1 + 2\sin x) dx + \int_0^{3\pi/2} \sin^2 x dx \right]$$

$$= \pi \left[x - 2\cos x \right]_0^{3\pi/2} + \pi \left(\frac{3\pi}{4} \right)$$

$$= \pi \left(\frac{3\pi}{2} + 2 \right) + \frac{3\pi^2}{4}$$

$$= \frac{3\pi^2}{2} + 2\pi + \frac{3\pi^2}{4}$$

$$= \left(\frac{9\pi^2}{4} + 2\pi \right) \text{ cubic units}$$

c) i) 15 students

$$\begin{aligned} \text{ways of arranging in circle} &= (15-1)! \\ &= 14! \end{aligned}$$

ii) Ways in which the 4 students don't sit together
 = total ways - no. of ways they sit together

No of ways they sit together:

$$12 \text{ groups} \Rightarrow (12-1)! \times 4!$$

$$\therefore \text{ways they sit apart} = 14! - 11! \cdot 4!$$

$$\text{Probability they sit apart} = \frac{14! - 11! \cdot 4!}{14!}$$

$$= \frac{11! (14 \times 13 \times 12 - 4!)}{14!}$$

$$= \frac{90}{91}$$

13 d) $P(x) = 4x^3 - 12x^2 + 5x + 6$

i) $P(2) = 4(2)^3 - 12(2)^2 + 5(2) + 6$
 $= 32 - 48 + 10 + 6$
 $= 0$

$\therefore (x-2)$ is a factor of $P(x)$

ii)

$$\begin{array}{r}
 \cdot 4x^2 - 4x - 3 \\
 x-2 \overline{) 4x^3 - 12x^2 + 5x + 6} \\
 \underline{- 4x^3 - 8x^2} \\
 -4x^2 + 5x \\
 \underline{- -4x^2 + 8x} \\
 -3x + 6 \\
 \underline{- -3x + 6} \\
 0
 \end{array}$$

$\therefore P(x) = (x-2)(4x^2 - 4x - 3)$

$= (x-2)(2x+1)(2x-3)$

(13)

Question 14 (10 marks)

$$a) \quad \underline{a} = \begin{pmatrix} -4 \\ 1 \end{pmatrix}, \quad \underline{b} = \begin{pmatrix} -3 \\ 5 \end{pmatrix}$$

$$ii) \quad |\underline{a}| = \sqrt{(-4)^2 + (1)^2} \\ = \sqrt{17}$$

$$ii) \quad \underline{a} \cdot \underline{b} = (-4) \times (-3) + (1) \times (5) \\ = 17$$

$$iii) \quad \underline{a} \cdot \underline{b} = |\underline{a}| \cdot |\underline{b}| \cos \theta$$

$$\cos \theta = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| \cdot |\underline{b}|}$$

$$|\underline{b}| = \sqrt{(-3)^2 + (5)^2} \\ = \sqrt{34}$$

$$= \frac{17}{\sqrt{17} \times \sqrt{34}}$$

$$\cos \theta = \frac{1}{\sqrt{2}}$$

$$\theta = \frac{\pi}{4}$$

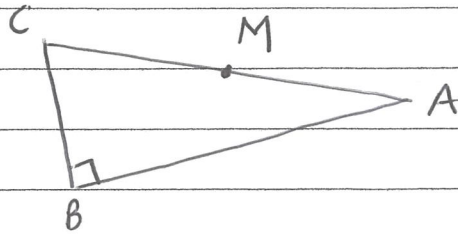
$$iv) \quad \text{proj}_{\underline{b}} \underline{a} = \frac{\underline{a} \cdot \underline{b}}{\underline{b} \cdot \underline{b}} \times \underline{b}$$

$$= \frac{17}{34} \begin{bmatrix} -3 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} -3/2 \\ 5/2 \end{bmatrix}$$

14 b)

$$\vec{AM} = \underline{a}, \quad \vec{BM} = \underline{b}$$



$$\begin{aligned} \text{i) } \vec{AB} &= \vec{AM} + \vec{MB} \\ &= \vec{AM} - \vec{BM} \\ &= \underline{a} - \underline{b} \end{aligned}$$

$$\begin{aligned} \vec{BC} &= \vec{BM} + \vec{MC} \\ &= \vec{BM} + \vec{AM} \\ &= \underline{b} + \underline{a} \\ &= \underline{a} + \underline{b} \end{aligned}$$

ii) $\triangle ABC$ is right angled

$$\therefore \vec{AB} \perp \vec{BC}$$

$$\Rightarrow \vec{AB} \cdot \vec{BC} = 0$$

$$\text{i.e. } (\underline{a} - \underline{b}) \cdot (\underline{a} + \underline{b}) = 0$$

$$\Rightarrow \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} - \underline{a} \cdot \underline{b} - \underline{b} \cdot \underline{b} = 0$$

$$|\underline{a}|^2 - |\underline{b}|^2 = 0$$

$$|\underline{a}| = |\underline{b}|$$

$$\therefore |\vec{AM}| = |\vec{BM}| = |\vec{CM}|$$

i.e. M is equidistant from the three vertices of $\triangle ABC$.

(15)

Question 15 (10 marks)

a) $T = 25 + 400e^{-0.05t}$, $t \geq 0$

i) T when $t=0$ (enters liquid)

$$T = 25 + 400e^0$$

$$= 425^\circ\text{C}$$

ii) $300 = 25 + 400e^{-0.05t}$

$$e^{-0.05t} = \frac{275}{400}$$

$$-0.05t = \ln\left(\frac{11}{16}\right)$$

$$t = 7.49 \text{ minutes (to 3 sig figs)}$$

iii) $T = 25 + 400e^{-0.05t}$

$$\frac{dT}{dt} = -20e^{-0.05t}$$

when $t=50$ $\frac{dT}{dt} = -20e^{-0.05(50)}$

$$= -1.64^\circ\text{C per minute}$$

iv) $T = 25 + 400e^{-0.05t}$

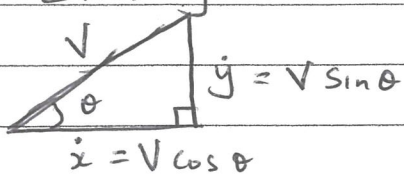
$$\text{As } t \rightarrow \infty, \frac{400}{e^{0.05t}} \rightarrow 0$$

$$\therefore T \rightarrow 25 > 20$$

$\therefore T$ never falls below 20°C

15 b) Initially

i)

Horizontally,

$$\ddot{x} = 0$$

$$\dot{x} = C_1$$

$$\text{When } t=0, \dot{x} = V \cos \theta$$

$$\therefore C_1 = V \cos \theta$$

$$\dot{x} = V \cos \theta$$

$$x = Vt \cos \theta + C_2$$

$$\text{When } t=0, x=0$$

$$\therefore C_2 = 0$$

$$x = Vt \cos \theta$$

Vertically

$$\ddot{y} = -10$$

$$\dot{y} = -10t + C_1$$

$$\text{When } t=0, \dot{y} = V \sin \theta$$

$$\therefore C_1 = V \sin \theta$$

$$\dot{y} = -10t + V \sin \theta$$

$$y = -5t^2 + Vt \sin \theta + C_2$$

$$\text{When } t=0, y=0$$

$$\therefore C_2 = 0$$

$$y = -5t^2 + Vt \sin \theta$$

ii) After 1.5 seconds, $x = 60$, $y = 2.25$

$$60 = 1.5 V \cos \theta ; 2.25 = -5(1.5)^2 + 1.5 V \sin \theta$$

$$V \cos \theta = 40 \quad \text{--- (1)}$$

$$V \sin \theta = 9 \quad \text{--- (2)}$$

$$\textcircled{1} \div \textcircled{2}$$

$$\frac{V \sin \theta}{V \cos \theta} = \frac{9}{40}$$

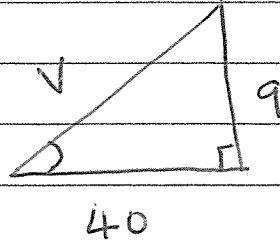
$$\tan \theta = \frac{9}{40}$$

$$\theta = 12^\circ 41'$$

The angle of projection is $12^\circ 41'$

15 b (iii) From ① and ② [Part (ii)]

$$V \sin \theta = 9 \quad ; \quad V \cos \theta = 40$$



$$V = \sqrt{9^2 + 40^2}$$

$$= 41 \text{ m/s}$$

∴ Initial speed of the ball
is 41 m/s.

End of Solutions